

t-SNE Project Warmup Exercise

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1 Setup

This document is dedicated to the reconstruction of [1] with low-dimension kernel g replaced by a more general kernel. This would serve as a great way to understand the material and help with the completion of the original project. For the sake of simplicity, we will use the exact same notation as in [1] and skip all the definitions. Recall in [1], g is defined as

$$g(y, y') = \frac{1}{1 + \|y - y'\|^2} \quad (1.1)$$

With all these being said, we will first start by giving our general low dimension kernel g some general assumptions, and modify it as we move on. We first require that $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' > \infty \quad (1.2)$$

We will kick off this warmup by listing all the important lemmas and results in order, and prove our own versions when the need arises. The final goal is to prove a same set of results as in [1] with the y kernel replaced. Recall that

$$\begin{aligned} F_{\rho, \mu}(x, \sigma) = & - \int \frac{\|x - x'\|^2}{2\sigma^2} \frac{\exp(-\|x - x'\|^2 / 2\sigma^2)}{\int \exp(-\|x - x'\|^2 / 2\sigma^2) d\mu(x')} d\mu(x') \\ & - \log \left(\int \exp(-\|x - x'\|^2 / 2\sigma^2) \right) + \log \rho. \end{aligned} \quad (1.3)$$

Proposition 1. *If μ has a C^1 density f such that both f and $\|\nabla f\|$ are bounded, then the equation $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho, \mu}^*(x)$ is smooth.*

Proof. This proposition is about the kernel of x . The proof stays unchanged from [1]. □

The next couple of lemmas are general results in section 3 of [1]. We will omit the proof because they are not directly discussing the quantities in the t-sne algorithm.

Lemma 1. *Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:*

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\delta \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\|x - x'\|/\sigma \rightarrow \infty$.
Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Let $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma \in K_n, x} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \quad (1.4)$$

Recall that X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2 \exp(-Ct^2)$ for some positive constant C .

Lemma 2. Let ν be sub-Gaussian with second moment equal to 1 and suppose $X \in \mathbb{R}^d$ has i.i.d. entries drawn from ν . Then there exists constant C, c_1 such that

$$\mathbb{P}(\|X\| \geq \sqrt{d} + t) \leq Ce^{-c_1 t^2}. \quad (1.5)$$

Define

$$f_\sigma(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2), \quad g_\sigma = \frac{\|x - x'\|^2}{2\sigma^2} \exp(-\|x - x'\|^2 / 2\sigma^2). \quad (1.6)$$

The next two lemmas use similar techniques to prove.

Lemma 3. Let c_1 be the decay constant in lemma 2 and let $\delta > 0$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu_n(x')} - \frac{\int g_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| = O(n^{8/c_1 \delta^2 - 1/2} \sqrt{\log n}). \quad (1.7)$$

Lemma 4. Let c_1 be the decay constant in lemma 2 and let $\delta > 0$ be such that $c_1 \delta^2 > 8$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \log \left(\int f_\sigma(X_i, x') d\mu_n(x') \right) - \log \left(\int f_\sigma(X_i, x') d\mu(x') \right) \right| = O(n^{4/c_1 \delta^2 - 1/2} \sqrt{\log n}) \quad (1.8)$$

For this pair of lemmas, [1] provides the proof to the first lemma. We will provide the second for the sake of exercise.

Proof of 4. I will just randomly type some words to see how this works. □

References

- [1] Antonio Auffinger and Daniel Fletcher. Equilibrium distributions for t-distributed stochastic neighbour embedding, 2023, 2304.03727. URL <https://arxiv.org/abs/2304.03727>.
- [2] Aditya Ravuri and Neil D. Lawrence. Towards one model for classical dimensionality reduction: A probabilistic perspective on umap and t-sne, 2024, 2405.17412. URL <https://arxiv.org/abs/2405.17412>.

- [3] Pavlin G. Poličar and Blaž Zupan. Visualizing high-dimensional temporal data using direction-aware t-sne, 2024, 2403.19040. URL <https://arxiv.org/abs/2403.19040>.
- [4] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9(2579-2605), 2008.
- [5] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.